

SPOKANE 17 SSW, WA, USA

WMO: 742040

Lat: 47.4174N

Lon: 117.5264W

Elev: 691

StdP: 93.29

Time Zone: -8.00 (NAP)

Period: 07-19

WBAN: 04136

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	-17.7	-13.7	-20.7	0.6	-16.7	-17.3	0.9	-12.2	5.6	5.1	5.0	4.1	0.2	N/A	0.547

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.9	33.8	16.6	32.0	16.3	30.1	15.8	18.3	29.8	17.4	29.4	16.6	28.4	1.9	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
14.5	11.2	21.8	13.0	10.2	19.9	11.9	9.4	19.3	54.2	29.8	51.2	29.3	48.7	28.3	22.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
4.8	4.2	3.8	-23.9	36.8	4.4	2.0	-27.1	38.2	-29.6	39.4	-32.1	40.5	-35.3	41.9		
			-22.6	19.8	4.0	1.3	-25.5	20.7	-27.9	21.5	-30.1	22.2	-33.1	23.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	7.3	-2.0	-0.7	3.2	6.7	11.4	14.4	18.2	17.8	12.8	6.7	1.5	-2.4
	DBStd	8.25	5.25	4.87	3.87	3.30	3.60	3.41	3.04	3.00	3.58	3.94	4.75	4.80
	HDD10.0	1791	373	301	210	108	27	3	0	0	13	117	256	383
	HDD18.3	4087	632	534	468	348	217	126	41	46	168	361	505	641
	CDD10.0	823	0	0	1	10	70	134	254	242	98	14	1	0
	CDD18.3	77	0	0	0	0	1	7	37	30	3	0	0	0
	CDH23.3	3306	0	0	0	8	86	323	1311	1236	331	10	0	0
	CDH26.7	1379	0	0	0	0	12	111	583	560	111	2	0	0
Wind	WSAvg	1.5	1.4	1.6	1.8	2.0	1.7	1.6	1.4	1.3	1.3	1.3	1.3	1.3
Precipitation	PrecAvg	409	55	36	35	32	37	30	11	11	13	36	53	61
	PrecMax	605	117	85	69	69	66	59	71	35	35	86	110	126
	PrecMin	297	32	5	7	12	8	5	0	0	0	7	13	27
	PrecStd	66	20	19	17	15	16	12	15	9	10	20	22	23
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	10.0	12.0	17.3	24.7	28.3	33.6	35.6	36.4	32.5	24.3	15.7	11.1
		MCWB	6.2	7.4	8.8	12.9	15.3	17.7	17.0	16.6	15.9	13.2	10.2	8.1
	2%	DB	7.4	9.5	14.3	20.6	25.7	30.0	33.6	34.0	29.9	20.5	12.8	8.0
		MCWB	4.7	6.1	7.8	11.1	14.4	16.8	16.7	16.5	15.0	11.4	9.1	5.8
	5%	DB	5.7	7.8	11.9	17.3	23.7	26.7	32.0	32.2	27.3	17.7	10.7	5.8
		MCWB	3.7	5.1	6.9	9.5	13.7	15.1	16.3	16.1	14.2	10.0	7.4	4.0
	10%	DB	4.2	6.0	10.0	14.6	21.2	24.5	30.3	30.0	24.6	15.0	8.9	4.3
		MCWB	2.6	3.8	6.0	8.0	12.2	14.2	16.1	15.4	13.2	8.9	6.4	2.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	6.8	8.0	10.5	14.7	16.4	19.3	20.2	18.4	17.9	14.5	11.3	8.1
		MCDB	9.2	11.3	14.5	23.4	25.4	32.0	31.0	32.2	27.9	21.0	14.7	10.4
	2%	WB	4.9	7.1	9.0	11.8	15.3	17.8	18.8	17.5	16.3	12.1	9.7	6.1
		MCDB	6.6	9.6	13.0	19.0	24.1	28.8	29.1	30.4	26.7	17.7	12.7	7.9
	5%	WB	3.6	5.8	7.9	10.3	14.4	16.4	17.6	16.8	14.9	11.1	8.1	4.4
		MCDB	5.1	7.7	11.8	17.2	22.7	26.4	29.6	29.9	26.3	16.1	10.3	5.9
	10%	WB	2.5	4.3	6.8	8.9	13.2	15.2	16.7	16.0	13.7	9.8	6.7	3.0
		MCDB	3.8	6.1	10.3	14.3	20.6	23.7	28.7	28.9	23.5	14.2	8.8	4.4
Mean Daily Temperature Range	5% DB	MDBR	7.5	9.0	11.1	12.8	15.2	15.9	20.9	21.5	20.0	15.3	10.0	7.4
		MCDBR	8.0	10.9	14.6	18.4	19.9	21.6	24.8	26.2	25.9	20.5	12.6	7.7
		MCWBR	6.4	7.3	9.1	9.8	9.6	9.6	10.0	10.7	12.3	12.1	8.3	6.1
	5% WB	MCDBR	7.9	8.9	12.7	17.3	18.0	20.4	21.5	23.1	22.3	16.5	10.8	7.0
		MCWBR	6.3	6.2	8.4	9.4	8.8	9.2	9.0	9.8	10.9	10.1	7.7	5.8
Clear-Sky Solar Irradiance	taub	0.270	0.278	0.291	0.316	0.329	0.330	0.330	0.339	0.316	0.301	0.279	0.268	
	taud	2.505	2.550	2.563	2.471	2.455	2.478	2.473	2.433	2.488	2.533	2.555	2.503	
	Ebn at Noon	837	909	943	941	935	932	927	905	899	863	821	795	
	Edh at Noon	61	73	84	102	107	106	105	104	90	73	58	55	
All-Sky Solar Radiation	RadAvg	1.07	2.00	3.07	4.58	5.87	6.36	7.42	6.16	4.40	2.62	1.26	0.84	
	RadStd	0.13	0.27	0.35	0.43	0.47	0.54	0.39	0.40	0.41	0.29	0.13	0.11	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (18 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+100	+47

Nomenclature: See separate page